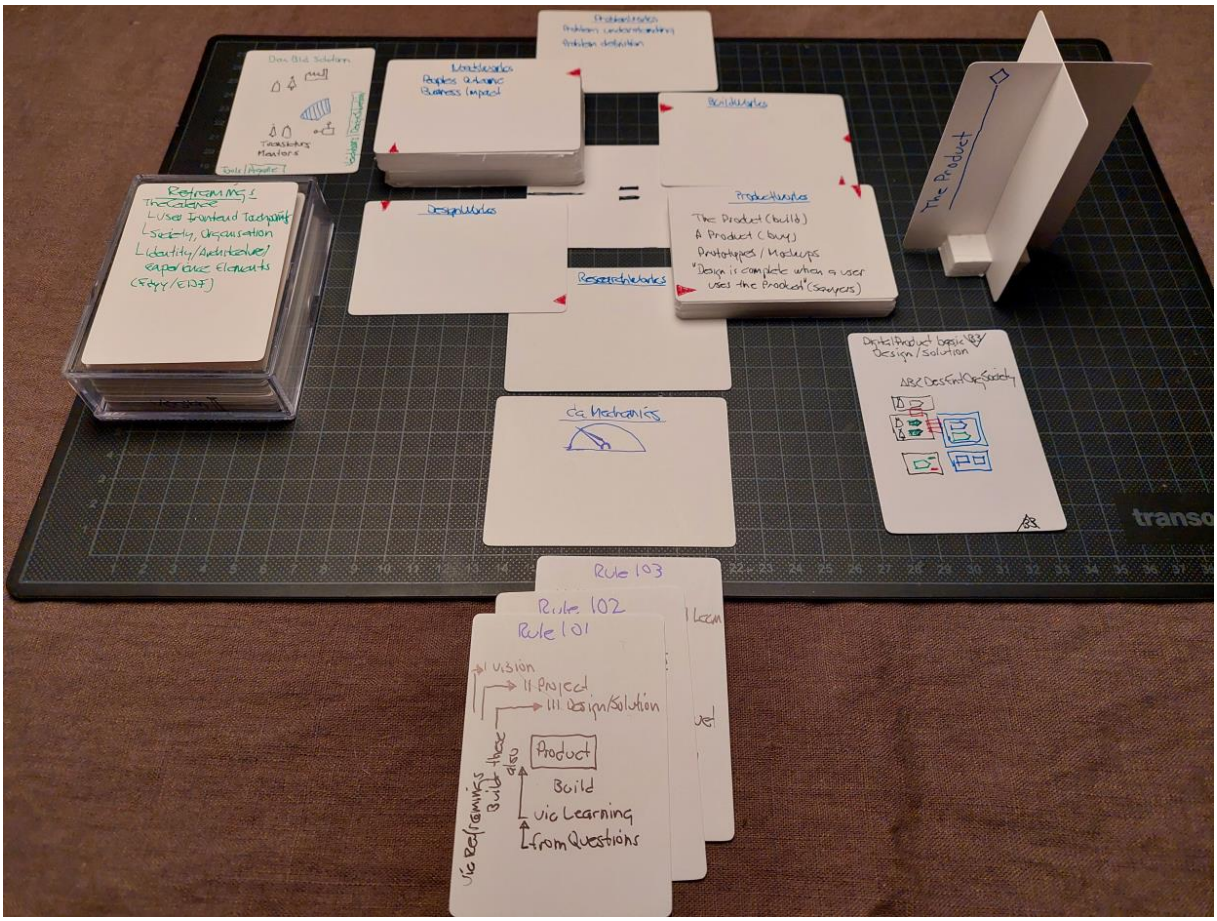
Product Requirements



This is a model of how to find out requirements, what customer and their user wants, their needs or 'user outcome' and what the business and their user and the organisation want, called 'business impact'.

The above shows all areas, all different main aspects that is needed to find this out. Experience shows that designers need to help customers discover what they want and that as designers, we can not discover this until watching users using the product and that the design of a product is not finished until users start using it. This is also in line with (Brooks 2010) and (Sayers 2006). Therefore, the model above does not only contain Problem and Need but also Design, how to Build and the Product.

This describes the Design Situation. Outside this there is a Project which is not shown but is explained in separate pattern

Needs and Requirements Work

This product describes a method and tools to discover problem and needs that user have on a digital product. The model above and description below describes how to do this according to our view. The basic principle is that design work needs to help customer discover what they need (Brooks 2010).

Each of these areas, as Problem, Need, Design, Build and Product-works are both areas where to put deliverables, it is areas to work on, it needs place and methods to perform.

The image above shows the 'design situation' with all needed parts. Outside this exist the project or line work where design is done, see pattern '_170 Plastic Box Design Situation' (tbd).

ProblemWorks

We always need to understand what the problem is. But this need not always be the first we do, the important thing is that we do. It can ultimately only be done 'while observing customer use the product'(Brooks 2010).

It is often described that Problem statements shall be more or less a oneliner, or it can be a 'Nyttoanalys' or something similar, quite brief. Instead, our experience is that problems instead is a vast collection of a lot of information. One pattern and tool for this is 'Periodical System' see pattern _120 'Problem Space (note that 'periodical' is tbd...) and accompanying templates in Visio and drawio for this work. It is also closely related to 'Research Work'.

NeedWorks

This is about specifying particular needs, users and what they want, the business and what that want. It might be in the form of use-cases, business-cases and as formal requirements. Normally test-cases are related to this area also. But again, the entire design situation or design space as shown above, are descriptions of what different needs are, only expressed as design, the product or how to build it.

DesignWorks, BuildWorks, ProductWorks

This is the space and place for designing and building a solution. Needs and requirements will appear here, specifically for the design, as constraints, non-functionals but also appear as what a user want, but found out during the design work. And, needs might as well be expressed in the design.

ProductWorks is a bit special. It is not the Product or solution, but the work about and around this, according to the thesis that users do not understand what they want until using the Product.

The '='-sign

The =-sign in the middle of the 'works' spaces is the basic methodology. Also referred to as 'Indigo' in other products and patterns. This is about the fact the development and design work is never done in steps, as most design and product development methodologies states. Instead Brooks(2010) amongst others describe design work as 'oscillating' around all the different works types in the design situation and finally, requirements and the product itself can only be seen as done when users start using it, also according to Sayers(2004) and illustrated with 'the Design Squiggle'(Damien 2025).

One way of visualizing this principle and way of working is by dyeing a piece of cloth, e.g. indigo-dyeing and thus the name. The cloth needs to be continuously pressed down, pressing out air all over the surfaces until the dye completely colors it.

Reframings

These are a stack of methods, frameworks used to get different perspectives, to be able to reframe the different works done. E.g. understanding the Problem might need to reframe from a technical, business and cost/revenue perspective etc.

The Object of design

This is the Product that shall be done. It is here represented with the 3D model to the right. Problem, Needs, a working Design in all aspects, how to Build can not be understood until observing the Customer or User using the Product (Brooks 2010) thus the design work is not done and the Product is not complete until used (Saywer 19nn).

The object of design is described by (Galle 2025), but this paper only identifies the problem and do not directly come with a really working solution. Anyway, the basic problem is that the Product begins with a 'thought in somebodys head'(Gabriel 1995) 'Mercy Street'.

Da Mechanics

Instead of following a process, this is where work is governed and to some extent managed. Management of work is mainly done in the adjoining project, see pattern '_170 Plastic Box Design Situation' (tbd) but this is any kind of project, the pattern just describes a generalized project, the important thing is how the design situation relates to the project, and separation of the two parts.

Instead this is basically a set of rules, implemented in various forms, with the aim to make sure the basic principles:

- Problem must be understood, but will not be fully understood until observing the User using the Product. Design works means to help Users and Customers discover what they really need.
- Needs need also be understood.
    - This is maybe the most underestimated and complex part which can also be visible in the model above. Needs work is central and depends on Problem but also Design, and Build and lastly the Product itself. It is at the start and also central.
    - To illustrate Needs which is also the subject for this product _100, the Needs area is slightly raised in the model above (put on foamboards to elevate above the design situation plane). This can also be viewed as it being the most complex part. In the model above, Product is also slightly raised, put on foamboard, to indicate the dependency between these two.
- Design needs work taking in all aspects, solving the Needs, using technology that is feasible and usable and design a solution that is desirable.

- Build must be possible and are often surprisingly complex in its own.
- Product then is both the actual Product and also all work around it so Product Works means also a box, manuals, distribution channels, support (note however that these later parts are more handled by project work).

To accompany da Mechanics, the reframings are used here to. As a simple example, assume we re frames towards business and cost/revenue to understand Problem and Need.

ABCDesEnt

This is a design pattern, see pattern _130.

This puts everything together and is thus a central and general pattern. It contains problem and needs from society, users/customers and employees in the organisation. It comprises process based analysis, which is predominant in many organisations but realised as capabilities which is a much more proper concept. It also reflects the basic business model of use-cases vs functionality and how the reconciliate, based upon the axiom that use cases are income, architecture is cost(Coplien 2010). It also is object oriented and thus reflects the underlying digital product design on a software level, both because it needs to be built this way but also involve the software design team in a natural way, making them understand from their perspective which is object models, how capabilities encapsulates and solves users needs.

The Blue Sector

This is both how the design situations is presented and how it involves different users, roles such as sponsors and stakeholders and domain experts but also end users and customers. Models for example, must always be adapted to a certain situation and role, such as a steering committee or presentation for a focus group, but still remain true to the design situation findings, direction, integrity.

PeopleWorks

Aka PeopleWare, PeopleWhere, PeopleWear

Projects some times just happens and the result is astounding. Projects sometimes just don't work at all. Projects are oftenmost just done, result is ok. Why, especially why some projects just becomes brilliant is a kind of mystery. To some part the complexity that we see elsewhere in systems work, but with people there is something else going on, emotions, communications, everyday life balance. This therefore deserves a space of its own in the design situation, it is very connected with the project and to some part therefore outside and with dependency on the design situation.

One problem with methods of various kind is that they often emanate from a successful project. Every method will be used in a certain combination of mission, team, organisation, resoursce and different people. And this special situation is impossible to copy even if we copy the method exactly. Even quite simple methods like Scrum is surprisingly hard to get to work really well.

John Cleese mentions this phenomenon in his book about creativity (Cleese 2018). There is a lot of other references and entire books written and in this work we will address as with all other tools, we try to point out what we have seen work, how to make it happen in real life design situations and together with other tools and methods. This is for now tbd, but Cleese book is worth reading as is 'Kreativitet'(Lohmann Stephensen 2018).

Specification and Form of Needs

Normally needs are specified as requirements. The same goes for a design, it is normally described in a design specifications. What we see from experience is that the form of this should be allowed to vary. What the model above shows is an equal sign '=' in between Need, Design, Build and Product works. This means that specifying a requirement, or a design can be done by showing a Product, where this can be in varying state of completeness, i.e. a mockup or prototype. This might be the most effective way to actually show what is needed, a real working product, maybe with additional description, in the simplest form a Post It identifying the need and pointing ut details.

Basic principle

Design processes often are described more or less as steps to follow, not always in sequence, there are iteration and other models. But instead, as Brooks notes, designers simply don't seems to work that way, and this is also our observation and reason we do this work where _Product 100 is the first attempt to something more practical and focus on needs and requirements. Instead, as Brooks notes, designers seems to oscillate over a certain design situation and what we try to do here, is instead to describe this design situation, what it consists of, how it relates to a project and the business driving the design. And also how the movement or navigation happens, with a set of rules and continuous discernment and judgement which is also what Design Way describes (Nelson, Stolterman 2014).

Other

The picture above shows the design situation. Requirements belongs to needs, what customers want. So this is highlighted in the model above. It is actually also raised above the plane, or elevated above the other parts of the design situation. But all parts are needed. E.g. 'DesignWork' also belongs to specifying requirements since it is needed to build the product and observe users using it and learn what they want. Design however is not in focus because the only design needed with focus on requirements, is to prepare a Product so that a user can use it, and it can be a mockup, or a simple prototype. In the full design work much other aspects are needed, all these aspects can be seen in the picture above as a stack of 're framings', to the left. E.g. non functional, software platform and other aspects need to be evaluated and decided upon, also BuildWorks are needed but is not in focus here.

But, to find out requirements we can do research, we can try do design or try to build and in each stage learn a lot. E.g. interviewing users is a good starting point, but as experience shows, the absolutely best way of learning what the requirements are, is not by asking a customer or user, but observing them, show them a product, encourage them to discover what it can do and learn

from a set of hypotheses we have set up, to be verified or falsified or by a more explorative, loose way of just observing the user struggle or succeed and learn from that, improve the product and show the user a new, updated version.